



POLITECNICO DI MILANO
SCHOOL OF INDUSTRIAL AND INFORMATION
ENGINEERING
BAYESIAN LEARNING AND MONTE CARLO SIMULATION

Final project:
Yacht Hydrodynamic Resistance

Authors:
Peng Rao (ID 270661)

Contents

1	Dataset Description	1
2	Log–log Regression	1
2.1	Model and prior	1
2.2	Posterior and Monte Carlo sampling	1
2.3	Results	2
	References	5
A	Tables	6

Abstract

This report presents a comprehensive study on the hydrodynamic resistance of yachts using Bayesian Learning and Monte Carlo Simulation techniques. The primary goal is to model and predict the resistance forces acting on yacht hulls as they move through water. By applying Bayesian inference methods, we estimate the posterior distributions of resistance coefficients based on observed data and prior knowledge. Monte Carlo simulation is employed to propagate uncertainty through the model, allowing for robust predictions that account for variabilities in environmental conditions and material properties. The results provide insights into the efficiency of different hull designs and contribute to the optimization of yacht performance in marine engineering applications.

1 Dataset Description

The data analysed in this report is the *Yacht Hydrodynamics* dataset from the UCI Machine Learning Repository [1].

The dataset comprises 22 different hull geometries, each tested at a range of speeds (Froude numbers), for a total of 308 observations (hull-form $\times$ Froude-number rows). Each observation is described by five hull-geometry parameters and the Froude number, together with the measured residuary resistance, which is the target variable. The variables are summarised in Table 1 in the appendix.

The first five covariates (`LCB`, `prismatic`, `L_disp`, `beam_draught`, `L_beam`) are geometric form factors that are constant within a given hull and characterise its shape. The Froude number is a dimensionless measure of speed and is the dominant predictor of resistance: for sailing yachts the residuary resistance is known to scale approximately as a power law of the Froude number, with an exponent close to 4 in the bulk regime. This physical expectation motivates the log–log regression and nonlinear modelling adopted in the remainder of the report. The target **resistance** is strongly right-skewed, ranging over more than three orders of magnitude, which makes a logarithmic treatment and an explicit account of heteroscedasticity essential for sound Bayesian inference.

2 Log–log Regression

2.1 Model and prior

The physics of slender sailing hulls suggests that, in the bulk regime, the residuary resistance grows as a power of the Froude number, resistance $\propto F n^{\gamma_F}$ with γ_F close to 4. Taking logarithms turns this multiplicative law into a linear model, in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(\text{Froude}) + \sum_{k=1}^5 \beta_k \text{hull}_k + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors (`LCB`, `prismatic`, `L_disp`, `beam_draught`, `L_beam`). Since the smallest observed resistance is $0.01 > 0$, the logarithm is well defined for every record and no offset is required. Writing the $n = 308$ observations in matrix form as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\beta} \in \mathbb{R}^p$, $p = 7$, we adopt the non-informative reference (Jeffreys) prior

$$p(\boldsymbol{\beta}, \sigma^2) \propto \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference and provides a transparent Bayesian baseline against which the model-selection and heteroscedasticity-corrected analyses of the following sections can be compared.

2.2 Posterior and Monte Carlo sampling

Under the prior (2) the posterior is Normal–Inverse–Gamma and is available in closed form. With $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ the least-squares estimate and

$s^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / (n - p)$ the unbiased error variance,

$$\sigma^2 \mid \mathbf{y} \sim \text{Inv-Gamma}\left(\frac{n-p}{2}, \frac{(n-p)s^2}{2}\right), \quad (3)$$

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}\left(\hat{\boldsymbol{\beta}}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right). \quad (4)$$

We sample the joint posterior by Monte Carlo: for each of $M = 20,000$ iterations we first draw σ^2 from (3) via a scaled inverse- χ^2 variate and then draw $\boldsymbol{\beta}$ from the Gaussian (4). The resulting sample $\{\boldsymbol{\beta}^{(m)}, \sigma^{2(m)}\}_{m=1}^M$ marginalises the regression coefficients over σ^2 (a multivariate Student- t) and is used to obtain all posterior means, standard deviations and 95% credible intervals, as well as the credible band of the fitted curve.

2.3 Results

Table 2 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale). The posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \quad 95\% \text{ CI} = [4.60, 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$. Fig. 1 (left) shows the log-log scatter with the posterior-mean line and its 95% credible band: the band is extremely tight, reflecting the 300 observations and the strong signal of the Froude number. The Monte Carlo posterior of β_F is displayed in Fig. 1 (right); the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \mathbf{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

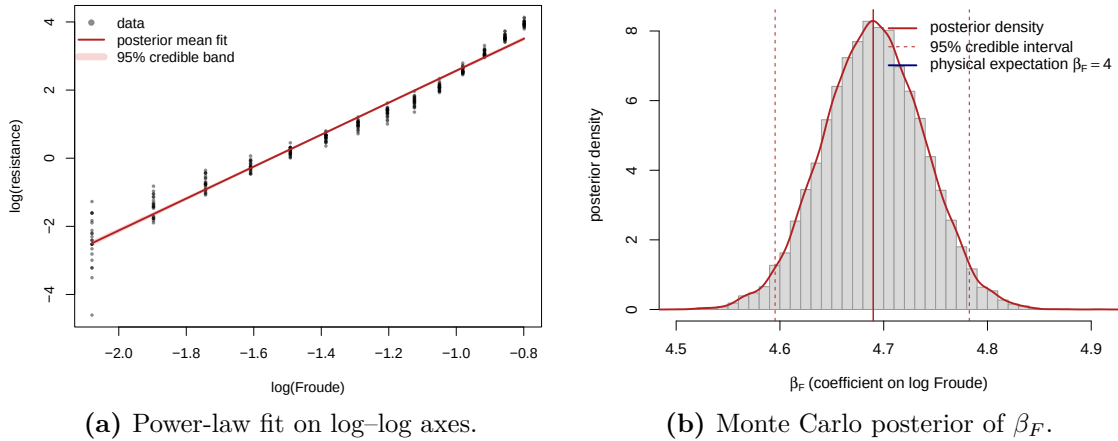


Figure 1: Left: observed $\log(\text{resistance})$ against $\log(\text{Froude})$ with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Monte Carlo posterior density of the Froude exponent β_F ; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval

comfortably contains zero (Table 2 and Fig. 3). The `prismatic` coefficient is especially poorly identified, with a posterior standard deviation of about 1.6, because the form factors vary only across the 22 hulls and are strongly collinear. This near-redundancy is precisely what the Bayesian model selection of the next section (BAS/BMA over the 2^5 submodels) is meant to resolve, by quantifying which covariates carry signal through their posterior inclusion probabilities rather than through single-model p -values.

Residual diagnostics. Although the global fit is excellent, the residuals reveal two systematic defects that the simple log-log model cannot capture (Fig. 2). First, the residuals-versus-fitted panel shows a clear U-shaped pattern: a single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds, consistent with the local Froude exponent increasing from ≈ 4.7 in the full range to above 6 in the high-speed subset. Second, the spread of the residuals is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 1. The Normal Q-Q plot confirms heavier-than-Gaussian lower tails. These two findings (speed-dependent curvature and non-constant variance) directly motivate the heteroscedasticity-corrected nonlinear model developed in the subsequent section.

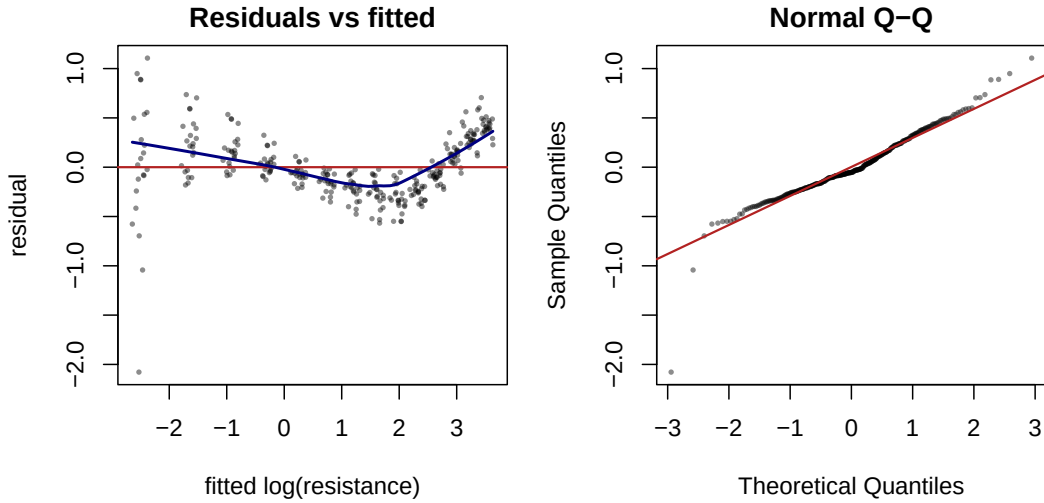


Figure 2: Residual diagnostics for the log-log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q-Q plot, showing heavier-than-Gaussian lower tails.

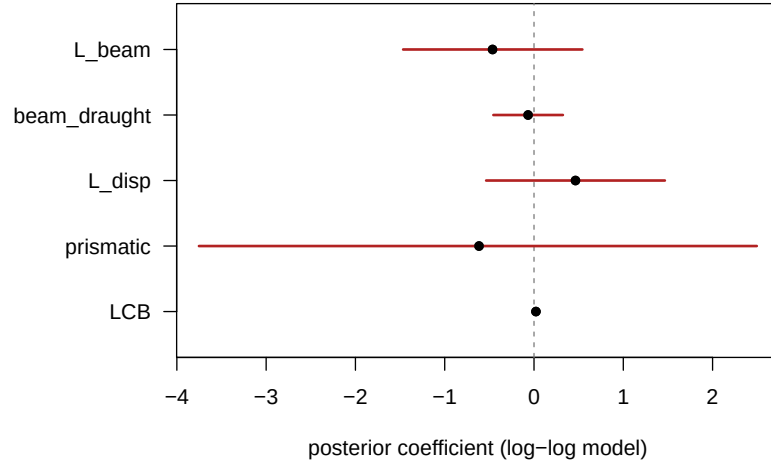


Figure 3: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log-log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

3 BAS

4 Heteroscedasticity-Corrected Nonlinear Model

References

- [1] I. Ortigosa, R. Lopez, and J. Garcia. *Yacht Hydrodynamics*. UCI Machine Learning Repository. DOI: 10.24432/C5XG7R. 2007.

A Tables

Table 1: Variables of the yacht hydrodynamics dataset.

Variable	Description
LCB	Longitudinal position of the centre of buoyancy
prismatic	Prismatic coefficient
L_disp	Length–displacement ratio
beam_draught	Beam–draught ratio
L_beam	Length–beam ratio
Froude	Froude number, $Fn = v/\sqrt{gL}$
resistance	Residuary resistance per unit weight of displacement

Table 2: Posterior summary of the log–log model (1) under the reference prior ($M = 20,000$ Monte Carlo draws). $\Pr(> 0)$ is the posterior probability that the coefficient is positive.

Coefficient	Mean	SD	2.5%	97.5%	$\Pr(> 0)$
Intercept β_0	7.183	0.977	5.245	9.122	1.00
log(Froude) β_F	4.690	0.048	4.595	4.782	1.00
LCB	0.021	0.012	−0.003	0.046	0.96
prismatic	−0.616	1.597	−3.753	2.495	0.35
L_disp	0.464	0.512	−0.536	1.466	0.82
beam_draught	−0.067	0.200	−0.455	0.324	0.37
L_beam	−0.464	0.515	−1.466	0.542	0.18